

Exercise 1

$$\left| \frac{V_4}{V_3} \right| = 0,6 \quad \angle \frac{V_4}{V_3} = \pi/8$$

$$E = 15V$$

$$Z_{n_i} = R_0 = 50 \Omega \quad \text{for } i = 1, 2, 3, 4$$

$$Z_g = R_0$$

$$Z_L = R_0$$

$$S = \begin{pmatrix} 0 & \sqrt{1-k^2} & jk & 0 \\ \sqrt{1-k^2} & 0 & 0 & jk \\ jk & 0 & 0 & \sqrt{1-k^2} \\ 0 & 0 & jk & 0 \end{pmatrix}$$

coupling factor
↑

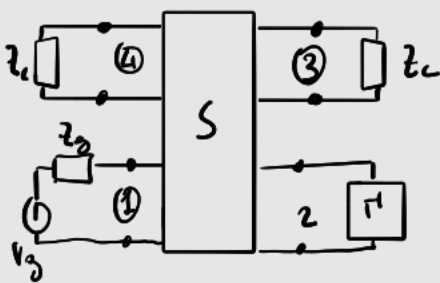
$$\text{with } -20 \log_{10} k = 3 \text{ dB}, k \in \mathbb{R}$$

$$C = -20 \log_{10}(k) \Rightarrow k = 10^{-\frac{3}{20}} = 0,7079$$

The through-path coefficient magnitude is:

$$\sqrt{1-k^2} = \sqrt{1-0,5012} = 0,7063$$

Let's analyze a bit the design in order to define a parameters



- The generator is connected at port 1, so the incident wave is a_1 . Since $Z_g = Z_{n_1}$, there are no secondary reflection from the generator.

- At port 2 there's the load, the incident wave at this port is $a_2 = \Gamma b_2$

- Ports 3 and 4 are matched with the reference impedance, so there are no incident wave from these ports $a_3 = a_4 = 0$

With the considerations above we can say that the reflected wave at port 2 is:

$$b_2 = \underbrace{S_{21}}_0 a_1 + \underbrace{S_{22}}_{\Gamma b_2} a_2 + \underbrace{S_{23}}_0 a_3 + \underbrace{S_{24}}_{jk} a_4 = S_{21} a_1 = a_1 \sqrt{1-k^2}$$

$$a_2 = \Gamma b_2 = \Gamma a_1 \sqrt{1-k^2}$$

Now we compute the outgoing waves at Port 3 and Port 4:

$$b_3 = S_{31} a_1 + S_{32} a_2 = jk a_1$$

$$b_4 = S_{41} a_1 + S_{42} a_2 = 0 + jk \Gamma a_1 \sqrt{1-k^2} = jk \Gamma \sqrt{1-k^2} a_1$$

Since these ports are matched $V_3 \propto b_3$ and $V_4 \propto b_4$ so the voltage ratio equals the wave ratio

$$\frac{V_4}{V_3} = \frac{b_4}{b_3} = \frac{jk \Gamma \sqrt{1-k^2} a_1}{jk a_1} = \Gamma \sqrt{1-k^2}$$

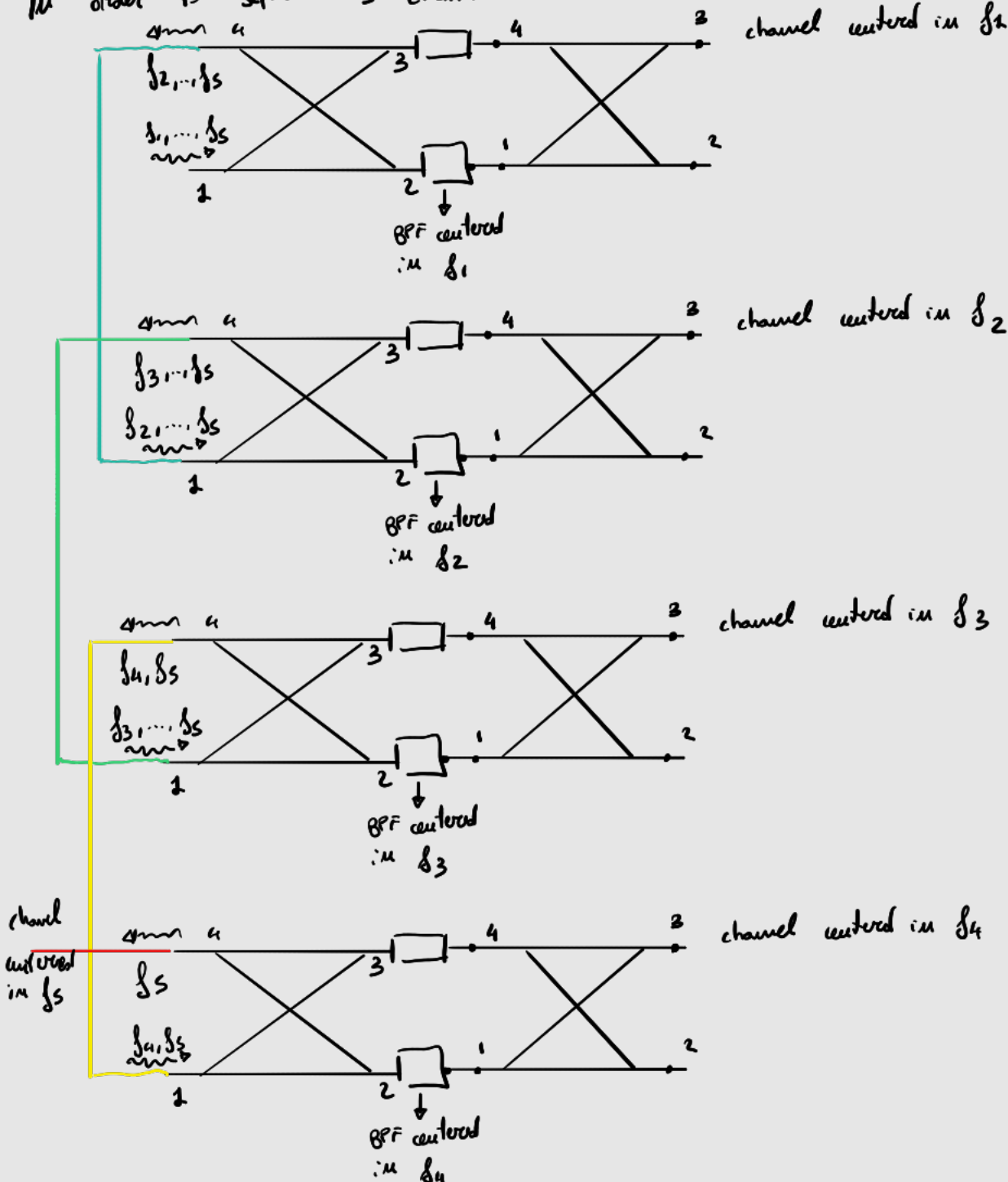
With this we can compute Γ

$$\frac{V_4}{V_3} = \left| \frac{V_4}{V_3} \right| \angle \left(\frac{V_4}{V_3} \right) = 0,6 e^{j\pi/8}$$

$$\frac{V_4}{V_3} = \Gamma \sqrt{1-k^2} \Rightarrow \Gamma = \frac{0,6 e^{j\pi/8}}{\sqrt{1-k^2}} = \frac{0,6}{0,7063} e^{j\pi/8} = 0,8495 e^{j\pi/8} = 0,7848 + 0,3251j \Omega$$

Exercise 2

In order to separate 5 channels we need 4 blocks arranged in this structure:



Now that we have the design clear we know that:

- Passing through a block multiplies by $|S_{21}|^2 = 0,95$
- Reflecting from a block, so moving to the next stage, multiplies by $|S_{11}|^2 = 0,97$

Now we evaluate the attenuation:

• $f_1 \rightarrow$ passes block 1

$$\frac{P_{out}}{P_{in}} = |S_{21}^f|^2 = 0,95$$

• $f_2 \rightarrow$ reflects block 1, passes block 2

$$\frac{P_{out}}{P_{in}} = |S_{11}^f|^2 |S_{21}^f|^2 = 0,97 \cdot 0,95 = 0,9215$$

• $f_3 \rightarrow$ reflects blocks 1 and 2, passes block 3

$$\frac{P_{out}}{P_{in}} = (|S_{11}^f|^2)^2 |S_{21}^f|^2 = 0,97^2 \cdot 0,95 = 0,8939$$

• $f_4 \rightarrow$ reflects blocks 1, 2 and 3, passes block 4

$$\frac{P_{out}}{P_{in}} = (|S_{11}^f|^2)^3 |S_{21}^f|^2 = 0,97^3 \cdot 0,95 = 0,8670$$

• $f_5 \rightarrow$ reflects all blocks

$$\frac{P_{out}}{P_{in}} = (|S_{11}^f|^2)^4 = 0,97^4 = 0,8353$$

The maximum attenuation is given by channel centered in f_5 since to get it we need to pass through most blocks

$$0,8670 \xrightarrow{\text{dB}} 10 \log_{10}(0,8670) = -0,62 \text{ dB}$$

$\hookrightarrow$ This is a power ratio!

Exercise 3

$$C = 3 \text{ dB} \pm 0,1 \text{ dB}$$

A system that does what's written in the pdf utilizes 4 90° hybrid directional couplers. The power transfer for a single coupler is either the coupled path or the through path.

We're given the coupling with a certain range, so we calculate their min and max

• Min $C = 3,1 \text{ dB} \rightarrow k^2 = 10^{-\frac{3,1}{10}} = 0,4898$

The through path is $1 - k^2 = 0,5102$

• Max $C = 2,9 \text{ dB} \rightarrow k^2 = 10^{-\frac{2,9}{10}} = 0,5129$

The through path is $1 - k^2 = 0,4871$

The worst case power transfer is when a signal must pass 2 times through the path with the minimum transmission coefficient.

$$\left. \frac{P_{out}}{P_{in}} \right|_{\min} = 0,4871 \cdot 0,4871 = 0,2373 \rightarrow \left| \frac{S_{out}}{S_{in}} \right|_{\min} = \sqrt{0,2373} = 0,4871$$

Exercise 4

- Coupler 1 (S'): input is P_{in} at port 1'. The coupled port 3' provides P_{out1} . The through port 2' connects to coupler 2.

$$P_{out1} = k_1^2 P_{in}$$

$$P_{in2} = (1 - k_1^2) P_{in}$$

- Coupler 2 (S''): the input P_{in2} enters in port 4'', the through port 3'' provides P_{out2} and the coupled port 2'' provides P_{out3} .

$$P_{out2} = (1 - k_2^2) P_{in2} = (1 - k_2^2)(1 - k_1^2) P_{in}$$

$$P_{out3} = k_2^2 P_{in2} = k_2^2 (1 - k_1^2) P_{in}$$

Since $P_{out1} = P_{out2} = P_{out3}$ then

$$k_1^2 = (1 - k_2^2)(1 - k_1^2) = k_2^2 (1 - k_1^2)$$

$$1 - k_2^2 = k_2^2 \Rightarrow 2k_2^2 = 1 \Rightarrow k_2 = \frac{1}{\sqrt{2}}$$

We substitute k_2^2 in the first expression

$$k_1^2 = 0,5 (1 - k_1^2) \Rightarrow 1,5 k_1^2 = 0,5 \Rightarrow k_1 = \frac{1}{\sqrt{3}}$$

Exercise 5

$$Z_0 = 50 \Omega$$

$$C_{re} = 15 \text{ dB}$$

$$f_0 = 5 \text{ GHz}$$

$$\epsilon_r = 10$$

$$h = 1,27 \text{ mm}$$

We find even and odd parameters for $C_{re} = 15 \text{ dB}$

$$k = 10^{-\frac{C_{re}}{20}} = 0,1778$$

$$Z_{re} = Z_0 \sqrt{\frac{1+k}{1-k}} = 50 \sqrt{\frac{1,1778}{0,8222}} = 59,84 \Omega$$

$$Z_{ro} = Z_0 \sqrt{\frac{1-k}{1+k}} = 50 \sqrt{\frac{0,8222}{1,1778}} = 41,77 \Omega$$

With these we interpolate on the graph and get

$$\frac{W}{h} \approx 0,8 \Rightarrow W = 0,8 \cdot 1,27 = 1,016 \text{ mm}$$

$$S/h \approx 0,4 \Rightarrow S = 0,4 \cdot 1,27 = 0,508 \text{ mm}$$

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + \frac{12h}{W}}} = \frac{10+1}{2} + \frac{10-1}{2} \frac{1}{\sqrt{1 + \frac{12}{0,8}}} = 6,625$$

$$l = \frac{\lambda_g}{4} = \frac{c}{480 \sqrt{\epsilon_{eff}}} \cdot \frac{3 \cdot 10^8}{4 \cdot 5 \cdot 10^9 \sqrt{6,625}} = 5,827 \text{ mm}$$

Now we start with the RTZ evaluation

$$V_p = \frac{c}{\sqrt{\epsilon_{eff}}} = \frac{3 \cdot 10^8}{\sqrt{6,625}} = 1,165 \cdot 10^8 \text{ m/s}$$

$$L \approx \frac{Z_{oe}}{V_p} = \frac{59,84}{1,165 \cdot 10^8} = 513,6 \text{ nH/m}$$

$$C \approx \frac{1}{Z_{oe} V_p} = \frac{1}{59,84 \cdot 1,165 \cdot 10^8} = 143,4 \text{ pF/m}$$

We assume L_m negligible, but we still have to compute C_m

$$Z_{o2} \approx \sqrt{\frac{L}{C + 2C_m}} \Rightarrow L + 2C_m = \frac{L}{Z_{o2}^2} \Rightarrow \frac{1}{2} \left(\frac{L}{Z_{o2}^2} - L \right) = 75,5 \text{ pF/m}$$

• Even Mode Analysis

$$\sin \theta \quad l = \frac{\lambda_g}{4} \Rightarrow 2kl = \pi \Rightarrow \Gamma'_{eA} = -\Gamma'_{eB}$$

$$\Gamma'_B = \frac{50 - 59,84}{50 + 59,84} = -0,0836 \Rightarrow \Gamma'_A = 0,0836$$

$$Z_{eA} = 59,84 \frac{1 + 0,0836}{1 - 0,0836} = 71,62 \Omega$$

$$V_{eA} = \frac{1}{\sqrt{2}} V_{g1} \left(\frac{71,62}{71,62 + 50} \right) = 0,4164 V_{g1}$$

$$I_{eA} = \frac{1}{\sqrt{2}} V_{g1} \left(\frac{1}{121,60} \right) = 0,00581 V_{g1}$$

• Odd mode Analysis

$$k_{odd} = \omega \sqrt{L(C + C_m)} = 386,5 \text{ rad}^{-1}$$

$$k_{odd} l = 386,5 \cdot 0,005827 = 2,251 \text{ rad}$$

$$\Gamma'_{oB} = \frac{50 - 41,77}{50 + 41,77} = 0,0897$$

$$\Gamma'_{oA} = 0,0897 e^{-2j 2,251} = -0,0187 + j 0,0877$$

$$Z_{oA} = 41,77 \frac{1 - 0,0187 + j 0,0877}{1 + 0,0187 - j 0,0877} = 39,63 + j 7,01 \Omega$$

$$V_{oA} = \frac{1}{\sqrt{2}} V_{g1} \left(\frac{39,63 + j 7,01}{89,63 + j 7,01} \right) = (0,3150 + j 0,0306) V_{g1}$$

$$I_{oA} = \frac{1}{\sqrt{2}} V_{31} \left(\frac{1}{89,63 + j7,01} \right) = (0,00784 - j0,00061) V_{31}$$

$$V_{c2}(A) = \frac{1}{\sqrt{2}} (V_{eA} - V_{oA}) = \frac{1}{\sqrt{2}} (0,4164 - 0,3150 - j0,0306) V_{31} = (0,0717 - j0,0216) V_{31}$$

$$V_{c1}(A) = \frac{1}{\sqrt{2}} (V_{eA} + V_{oA}) = (0,5172 + j0,0216) V_{31}$$

$$I_{c1}(A) = \frac{1}{\sqrt{2}} (I_{eA} + I_{oA}) = (0,00965 - j0,00043) V_{31}$$

The incident voltage at port 1 is:

$$V_{c1}^+(A) = \frac{1}{2} (V_{c1}(A) + 50 I_{c1}(A)) \approx 0,5 V_{31}$$

S_{31} , the coupling is:

$$S_{31} = \frac{V_{c2}(A)}{V_{c1}^+(A)} = \frac{0,0717 - j0,0216}{0,5} = 0,1434 - j0,0432$$

$$|S_{31}| = 0,1497$$

$$C_{dc} = -20 \log_{10}(0,1497) = 16,5 \text{ dB.}$$